

Parameters: $300 \text{ lb}_f < F < 500 \text{ lb}_f$

$$e_{\max} = \frac{FL}{AE} = 0.009 \text{ in}$$

Aluminum $E_A = 8.5 \times 10^6 \text{ to } 11.5 \times 10^6 \text{ psi}$

$$F = 400 \text{ lb} \quad E_A = 8.5 \times 10^6 \text{ psi} \quad A = 4 \text{ in}^2$$

$$0.009 \leq \frac{(400 \text{ lb})(L \text{ in})}{(4 \text{ in}^2)(8.5 \times 10^6 \text{ psi})}$$

$$L \geq 765 \text{ in}$$